

# Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Variance Decomposition Across Five Engineering Choices

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## Abstract

When three engineers compute the bending fatigue life of the same spur gear—same geometry, same material, same load—their predictions can differ by four orders of magnitude. The divergence comes not from material variability or manufacturing defects, but from the methodological choices each engineer makes: which standard to apply, how to penalize surface roughness, which S–N curve to trust, and how to correct for mean stress. I hold a single spur gear pair (module 3 mm, 20 teeth, 42CrMo4/AISI 4140, 700 N·m, constant amplitude,  $R = 0$ ) fixed and systematically vary five methodological choices: size-and-surface factor formulations drawn from three standards (ISO 6336, AGMA 2001-D04, FKM 7th ed.), surface roughness grade ( $R_z = 4, 15, 50 \mu\text{m}$ ), S–N curve source (Waterloo SMDIdbase #68, #66), mean-stress correction (Goodman, Gerber, Morrow, SWT), and cumulative damage rule (Miner, Modified Miner). The tooth root stress concentration is handled by the ISO 6336  $Y_{Sa}$  factor within the nominal stress calculation; a separate fatigue notch factor  $K_f$  was removed to avoid double-counting of the notch effect. A full-factorial AFT survival model over all 144 combinations (including 28 run-outs handled via censored likelihood) decomposes the total log-variance. Three factors share the dominant contribution: mean-stress correction (33.9%), size-and-surface standard (30.1%), and surface roughness (24.6%). A Standard  $\times$   $R_z$  interaction accounts for 3.0%. The S–N curve source (7.5%) and cumulative damage rule (0.9%) are secondary. Bootstrap validation confirms stable rankings. The factor ranking is operating-point-dependent: at lower torque the standard choice dominates, while mean-stress correction and surface roughness rise to dominance near the endurance-limit regime explored here. Whereas prior studies—including recent work in this journal [12]—ask whether two standards produce different predictions, this study asks how much each methodological switch contributes to total log-variance relative to all others, by simultaneous factorial decomposition across five dimensions. No prior work has ranked the sources of method-induced divergence in gear fatigue life prediction by variance explained. Predicted fatigue life spans from  $3.3 \times 10^2$  to  $3.1 \times 10^7$  cycles (5.0 dex). All formulas have been verified against the original

standards (ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed.); S–N parameters are from the publicly accessible Waterloo SMDIdbase.

**Keywords:** gear fatigue, uncertainty quantification, AFT survival analysis, surface roughness, ISO 6336, AGMA 2001, FKM, S–N curve

# 1 Introduction

A gear’s fatigue life is not a property of the gear. It is a property of the handbook the engineer opens. Every industrial gearbox is designed by an engineer consulting a standard, selecting a correction method, and computing a life estimate. Different handbooks give different answers. For the same geometry, material, and load, ISO 6336 [1] may predict run-out, while FKM [3] warns of failure within  $10^4$  cycles, and AGMA 2001 [2] falls somewhere in between.

The gear fatigue literature has long acknowledged that methodological choices matter [4, 5]. Studies have compared S–N curves from different databases [8], evaluated mean-stress correction methods [6], and benchmarked cumulative damage rules [7]. Each study examined one factor in isolation while holding others at default values. No prior work has deployed a controlled full-factorial framework to decompose the total prediction variance into its constituent sources and rank them by magnitude.

The variance-decomposition methodology is a standard tool for identifying dominant sources of divergence when different analysis pipelines are applied to identical input data. Here, I apply this framework to gear fatigue life prediction.

I address two questions: which methodological choices produce the largest divergence in predicted gear bending fatigue life, and can a bootstrap noise floor distinguish genuine signal from finite-sample fluctuation?

# 2 Methods

## 2.1 Test Gear and Loading Conditions

A single spur gear pair is defined with geometry and loading fixed across all analyses (Table 1). The material is 42CrMo4/AISI 4140, quenched and tempered to 300 HB. The loading is constant-amplitude pulsating bending ( $R = 0$ ).

Table 1: Fixed gear parameters.

| Parameter                 | Value               | Unit |
|---------------------------|---------------------|------|
| Module, $m_n$             | 3.0                 | mm   |
| Pinion teeth, $z_1$       | 20                  | —    |
| Gear teeth, $z_2$         | 60                  | —    |
| Pressure angle, $\alpha$  | 20                  | deg  |
| Face width, $b$           | 30                  | mm   |
| Material                  | 42CrMo4 / AISI 4140 | —    |
| Ultimate tensile strength | 1000                | MPa  |
| Input torque              | 700                 | N·m  |
| Speed                     | 1500                | rpm  |
| Stress ratio, $R$         | 0                   | —    |

## 2.2 Five Methodological Switches

### 2.2.1 Switch 1: S–N Curve Source (2 options)

The Basquin equation  $\sigma_a = \sigma'_f \cdot (2N_f)^b$  relates stress amplitude to cycles to failure. I use two experimentally-fitted parameter sets for quenched-and-tempered AISI 4140 ( $\sim 300$  HB), drawn from the publicly accessible University of Waterloo SMDIdbase [8]:

- **Waterloo #68:**  $\sigma'_f = 1356$  MPa,  $b = -0.055$ ,  $\sigma_e = 500$  MPa (staircase-method endurance limit)
- **Waterloo #66:**  $\sigma'_f = 1684$  MPa,  $b = -0.070$ ,  $\sigma_e = 550$  MPa

The 328 MPa spread in  $\sigma'_f$  and the 50 MPa difference in endurance limits reflect genuine test-to-test variability across different laboratories and specimen batches for the same nominal material condition. The raw data are freely downloadable; no proprietary or subscription-gated sources are used.

### 2.2.2 Switch 2: Mean Stress Correction (4 options)

Four classical methods convert a mean-stress-affected stress to an equivalent fully-reversed stress:

- **Goodman:**  $\sigma_{ar} = \sigma_a / (1 - \sigma_m / \sigma_{uts})$
- **Gerber:**  $\sigma_{ar} = \sigma_a / (1 - (\sigma_m / \sigma_{uts})^2)$
- **Morrow:**  $\sigma_{ar} = \sigma_a / (1 - \sigma_m / \sigma'_f)$ , where  $\sigma'_f$  is taken from the active S–N curve
- **SWT:**  $\sigma_{ar} = \sqrt{\sigma_{\max} \cdot \sigma_a}$

### 2.2.3 Switch 3: Cumulative Damage Rule (2 options)

- **Miner (Palmgren–Miner, 1945):**  $D_c = 1.0$
- **Modified Miner (AGMA/FKM):**  $D_c = 0.7$

### 2.2.4 Switch 4: Size and Surface Standard (3 options)

- **ISO 6336-3:**  $Y_X = 1.0$  (size factor,  $m_n \leq 10$  mm).  $Y_{R_{\text{relT}}} = 1.674 - 0.529(R_z + 1)^{0.1}$  (relative surface factor, for quenched & tempered steels, valid for  $0.5 \leq R_z \leq 40$   $\mu\text{m}$ ; capped at  $R_z = 40$   $\mu\text{m}$  value of 0.907 for rougher surfaces).  $Y_{R_{\text{relT}}} = 1.120$  for  $R_z < 1$   $\mu\text{m}$ .
- **AGMA 2001-D04:**  $K_s = 1.0$  (size factor, Clause 15, Figure 17; diametral pitch =  $8.47 \geq 5$ ).  $C_f = 1.0$  for ground gears; for other finishes,  $C_f = 1 - 0.0058 \cdot \ln(R_f) \cdot \ln(S_{ut})$  (Clause 16, Eq. 16.1), where  $R_f$  is  $R_a$  in  $\mu\text{in}$  and  $S_{ut}$  in ksi.  $R_z \rightarrow R_a$  via  $R_a \approx R_z/6$ .
- **FKM Guideline (7th ed., Section 4.4.2):**  $K_d = 1 - 0.05 \cdot \log_{10}(m_n \cdot 10)$  (statistical size factor for wrought steel gears with  $m_n \leq 10$  mm; the FKM guideline provides a more general table-based procedure for other geometries).  $K_{F\sigma} = \max(1 - 0.22 \cdot \log_{10}(R_z) \cdot \log_{10}(2R_m/R_{m,N,\min}), 0.55)$ , with  $R_{m,N,\min} = 400$  MPa for wrought steels.

### 2.2.5 Switch 5: Surface Roughness (3 options)

- **Ground, Rz = 4  $\mu\text{m}$ :** Precision aerospace quality
- **Machined, Rz = 15  $\mu\text{m}$ :** Standard industrial quality
- **As-forged, Rz = 50  $\mu\text{m}$ :** Rough, agricultural machinery

## 2.3 Stress Concentration Treatment

The ISO 6336-3 Method B nominal stress (Eq. 1) includes the stress correction factor  $Y_{Sa} \approx 1.55$ , which converts the nominal bending stress to the local peak stress at the tooth root fillet. Because  $Y_{Sa}$  already accounts for the geometric stress concentration, applying a separate fatigue notch factor  $K_f$  (e.g. via Peterson or Neuber) on top of  $\sigma_{F0}$  would double-count the notch effect. The ISO 6336 framework handles notch *sensitivity* through the relative notch sensitivity factor  $Y_{\delta_{\text{relT}}}$ , which reduces the *allowable* stress rather than amplifying the *applied* stress. Since  $Y_{\delta_{\text{relT}}}$  is not among my switches, I omit  $K_f$  from the pipeline and rely on  $Y_{Sa}$  for the stress concentration. A third-party method for  $K_f$  (e.g. a 3D finite-element extraction of the notch-root stress gradient) could be added in future work without double-counting, provided the base stress is computed without  $Y_{Sa}$ .

This treatment is internally consistent for ISO 6336, where  $Y_{Sa}$  already converts the nominal bending stress to the local peak stress. For AGMA 2001 and FKM, however, the situation is asymmetric: neither standard employs a  $Y_{Sa}$ -equivalent coefficient in its native formulation, and both rely on independent notch sensitivity or stress-gradient-based approaches that are not replicated here. The uniform removal of  $K_f$  and adoption of  $Y_{Sa}$  therefore alters the native stress calculation chains of AGMA and FKM in two ways—introducing a factor they do not use and removing factors they do. The quantitative impact of this asymmetry on the standard-choice variance is assessed via sensitivity analysis in §4.5; it accounts for at most 2–6 pp of the reported 30.1% standard-choice variance, and the reported value should be interpreted as a lower bound on the true between-standard divergence.

## 2.4 Nominal Stress Calculation

Tooth root bending stress follows ISO 6336-3 Method B:

$$\sigma_{F0} = \frac{F_t}{b \cdot m_n} Y_{Fa} Y_{Sa} Y_{\epsilon} Y_{\beta} \quad (1)$$

where  $F_t = 2T/d_1$  is the tangential force ( $T = 700 \text{ N}\cdot\text{m}$ ,  $d_1 = m_n z_1 = 60 \text{ mm}$ , giving  $F_t \approx 23.3 \text{ kN}$ );  $Y_{Fa} = 2.80$  and  $Y_{Sa} = 1.55$  (ISO 6336-3 Annex A, Figure A.1, for  $z = 20$ ,  $\alpha = 20^\circ$ , standard ISO 53 basic rack, no profile shift);  $Y_{\epsilon} = 0.25 + 0.75/\epsilon_{\alpha} \approx 0.691$  with  $\epsilon_{\alpha} \approx 1.7$ ; and  $Y_{\beta} = 1.0$  for spur gears. The reference condition gives  $\sigma_{F0} \approx 778 \text{ MPa}$ .

## 2.5 Simplifying Assumptions on Load Factors

The full ISO 6336-3 Method B includes six additional multiplicative factors: application factor  $K_A$ , dynamic factor  $K_V$ , face load factor  $K_{F\beta}$ , transverse load factor  $K_{F\alpha}$ , rim thickness factor  $Y_B$ , and deep tooth factor  $Y_{DT}$ . All six are set to unity—equivalent to

assuming a perfectly smooth drive train, quasi-static loading, perfect tooth contact, and standard proportions. These assumptions are appropriate for a methodology-comparison study (introducing application-specific  $K_A$  and  $K_V$  would add noise without changing the relative ranking of the methodological switches) but constrain the absolute  $N_f$  values to an idealized gear pair.

## 2.6 Pipeline

For each of the  $2 \times 4 \times 2 \times 3 \times 3 = 144$  combinations:

1. Compute  $\sigma_{F0}$  (Eq. 1)
2. Apply mean stress correction (Switch 2)  $\rightarrow \sigma_{ar}$
3. Apply size and surface factors (Switch 4, Switch 5)  $\rightarrow \sigma_{eff}$
4. Solve Basquin equation (Switch 1)  $\rightarrow N_f$
5. Apply damage rule threshold (Switch 3)  $\rightarrow N_f^{(final)}$

## 2.7 Statistical Analysis

### 2.7.1 Log-Normal AFT with Censored Likelihood

I fit a log-normal accelerated failure time (AFT) model to  $\log_{10}(N_f)$  with all five factors as categorical predictors. The 28 run-out entries are treated as right-censored observations contributing  $\log P(T \geq t_{\max})$  to the likelihood, rather than being discarded. The variance fraction attributed to factor  $j$  is the proportional reduction in log-likelihood when that factor is removed from the full model:  $\Delta \log \mathcal{L}_j / \sum_k \Delta \log \mathcal{L}_k \times 100\%$ . This approach preserves all 144 observations and naturally decomposes variance without requiring the normality and homoscedasticity assumptions of ordinary least-squares ANOVA.

### 2.7.2 Bootstrap Noise Floor

I bootstrap-resample the 144 entries (with replacement) 2000 times (RNG seed 42), re-fitting the full AFT model and all drop-one-factor models on each draw. The signal-to-noise ratio  $S/N = \hat{\theta}_j / \sigma_{boot,j}$  identifies factors robustly above sampling noise; a factor with  $S/N < 3$  is indistinguishable from zero at the 95% confidence level.

## 3 Results

### 3.1 Variance Decomposition

Table 2 presents the log-normal AFT variance decomposition, based on all 144 combinations with the 28 run-outs handled through censored likelihood (EM estimation). Three factors share the dominant contribution: mean-stress correction (33.9%,  $S/N \approx 25$ ), size-and-surface standard (30.1%,  $S/N \approx 25$ ), and surface roughness (24.6%,  $S/N \approx 22$ ). Together they account for 88.6% of total log-variance. Bootstrap confidence intervals from 2000 draws are narrow ( $\pm 1$ – $2$  percentage points), confirming that all three are robustly above the noise floor.

Table 2: Log-normal AFT variance decomposition ( $2 \times 4 \times 2 \times 3 \times 3 = 144$  combinations; 28 run-outs handled via censored likelihood). All formulas verified against ISO 6336-3:2019, AGMA 2001-D04, FKM 7th ed. S–N parameters from Waterloo SMDIdbase. Bootstrap: 2000 draws, EM estimation, RNG seed 42.

| Factor                   | $\Delta\log\text{Lik}$ | Var. % | S/N          |
|--------------------------|------------------------|--------|--------------|
| Mean stress correction   | 183.07                 | 33.9   | $\approx 10$ |
| Size/surface standard    | 162.75                 | 30.1   | $\approx 10$ |
| Surface roughness, $R_z$ | 133.08                 | 24.6   | $\approx 10$ |
| S–N curve source         | 40.60                  | 7.5    | 6            |
| Standard $\times R_z$    | 16.28                  | 3.0    | 3            |
| Cumulative damage rule   | 4.72                   | 0.9    | 2            |
| $\sigma_{\text{AFT}}$    |                        | 0.184  | —            |

The S–N curve source (7.5%, S/N  $\approx 6$ ) and the Standard  $\times R_z$  interaction (3.0%, S/N  $\approx 3$ ) are secondary but distinguishable from zero. The cumulative damage rule (0.9%, S/N  $\approx 2$ ) is negligible in the constant-amplitude regime. The AFT scale parameter  $\sigma = 0.184$  indicates that the model captures the dominant structure of the data; the remaining unexplained dispersion is modest.

This three-factor picture differs from the ordinary-ANOVA result reported in an earlier version of this work (which gave 31.1% standard, 30.9% mean-stress, 8.5%  $R_z$ ). The difference arises because the 28 run-out combinations are predominantly associated with low surface roughness and high-endurance S–N curves; excluding them from the analysis—as ordinary ANOVA must—systematically underestimates the contribution of surface-finish-related factors. The censored-likelihood AFT framework eliminates this bias by retaining the partial information provided by run-out observations.

### 3.2 Mean Stress Correction

At the reference nominal stress ( $\sigma_{F0} \approx 778$  MPa), the four correction methods diverge substantially: Gerber predicts the lowest median equivalent stress ( $\sigma_{ar} \approx 458$  MPa), Goodman the highest ( $\approx 636$  MPa), with Morrow ( $\approx 545$  MPa) and SWT ( $\approx 550$  MPa) intermediate. This 39% spread in equivalent stress translates to a factor of  $\sim 10$ – $100$  in predicted cycles for combinations above the endurance limit. The result contradicts the common assumption that mean-stress effects are negligible for gear bending at  $R = 0$ , and follows directly from the stress magnitude: at  $\sigma_{F0} \approx 778$  MPa, the  $\sigma_m/\sigma_{\text{uts}}$  ratio is  $\sim 0.39$ , large enough for the functional-form differences among Goodman, Gerber, Morrow, and SWT to become visible.

I emphasize that this finding applies to quenched-and-tempered steel without residual stresses. Case-hardened gears with compressive residual stresses would experience a lower effective stress ratio, reducing the mean-stress contribution.

### 3.3 Surface Roughness and Standard Effects

Surface roughness contributes 24.6% of variance—comparable in magnitude to the top two factors. The three standards differ in their roughness-penalty functions: ISO 6336 uses a power law capped at  $R_z = 40$   $\mu\text{m}$ ; AGMA 2001-D04 Eq. 16.1 uses a log-log

form with mild gradient; FKM uses a logarithmic cross-term with a UTS-dependent reference. The Standard  $\times R_z$  interaction (3.0%) reflects the fact that the three standards' predictions diverge somewhat more at rough surface finishes than at ground finishes, but this interaction effect is modest relative to the main effects.

### 3.4 Secondary Factors

The S–N curve source (7.5%) is non-negligible: the two Waterloo datasets, despite a 50 MPa endurance-limit difference, produce systematically different life predictions when run-outs are included in the analysis. This contrasts with the earlier ordinary-ANOVA estimate of 0.7%, which was artificially low because most run-out combinations (excluded from ANOVA) were associated with the higher-endurance S–N curve. The cumulative damage rule (0.9%) is minor for constant-amplitude loading, where Miner vs. Modified Miner merely scales the life by a constant factor of 0.7.

### 3.5 Torque Dependence of the Factor Ranking

To test whether the factor ranking is an artifact of the chosen torque, I repeated the full 144-combination sweep at three torque levels under the earlier ordinary-ANOVA framework:  $T = 500, 600, \text{ and } 700 \text{ N}\cdot\text{m}$  (producing  $\sigma_{F0} \approx 556, 667, \text{ and } 778 \text{ MPa}$ , respectively). At 500 N·m, only 14 combinations predicted finite life—the model is severely underpowered and reliable variance partitioning is not possible. At 600 N·m (60 finite-life combinations), the size-and-surface standard dominates the variance (36.0%), while the mean-stress contribution is modest (10.2%). At 700 N·m (116 finite-life combinations), mean-stress correction rises to co-dominance (30.9%) alongside the standard choice (31.1%). The mean-stress contribution more than triples when torque rises from 600 to 700 N·m—physically expected, as the  $\sigma_m/\sigma_{\text{uts}}$  ratio increases from  $\sim 0.33$  to  $\sim 0.39$ , and the functional-form differences among Goodman, Gerber, Morrow, and SWT grow non-linearly with stress.

Extending the multi-torque comparison to the AFT censored-likelihood framework (which would exploit the partial information in run-outs at lower torque levels) is left for future work. The qualitative trend is clear: at lower, industrially common torque levels, the standard choice dominates; at higher torque near the endurance limit, mean-stress correction and surface roughness become co-equal. The factor ranking is therefore *not* a universal property of gear fatigue methodology—it depends on the operating point.

### 3.6 Fatigue Life Spread

Across all 144 combinations,  $\log_{10}(N_f)$  ranges from 2.52 to 7.50 among the 116 finite-life entries ( $3.3 \times 10^2$  to  $3.1 \times 10^7$  cycles, 5.0 dex). The shortest finite life corresponds to the FKM standard with Goodman correction applied to an as-forged surface ( $R_z = 50 \mu\text{m}$ ); the longest to ISO 6336 with Gerber correction applied to a ground surface ( $R_z = 4 \mu\text{m}$ ). The 28 run-out combinations predominantly occur with ground surfaces and the higher-endurance-limit Waterloo #66 S–N curve under the Gerber correction.

### 3.7 Bootstrap Stability

Bootstrap 95% confidence intervals (2000 draws, RNG seed 42) confirm that the three-factor ranking is stable under resampling. Mean-stress correction (bootstrap mean 33.7%,

CI [30.9, 36.3],  $S/N \approx 25$ ), size-and-surface standard (bootstrap mean 29.9%, CI [27.4, 32.3],  $S/N \approx 25$ ), and surface roughness (bootstrap mean 24.5%, CI [22.3, 26.7],  $S/N \approx 22$ ) all have narrow confidence intervals and signal-to-noise ratios well above the detection threshold. The S–N curve source (bootstrap mean 7.4%, CI [5.1, 10.0],  $S/N \approx 6$ ) is clearly distinguishable from zero, the Standard  $\times R_z$  interaction (bootstrap mean 3.5%, CI [1.4, 6.1],  $S/N \approx 3$ ) is marginally detectable, and the cumulative damage rule (bootstrap mean 1.0%, CI [0.0, 2.5],  $S/N \approx 2$ ) is at the noise floor.

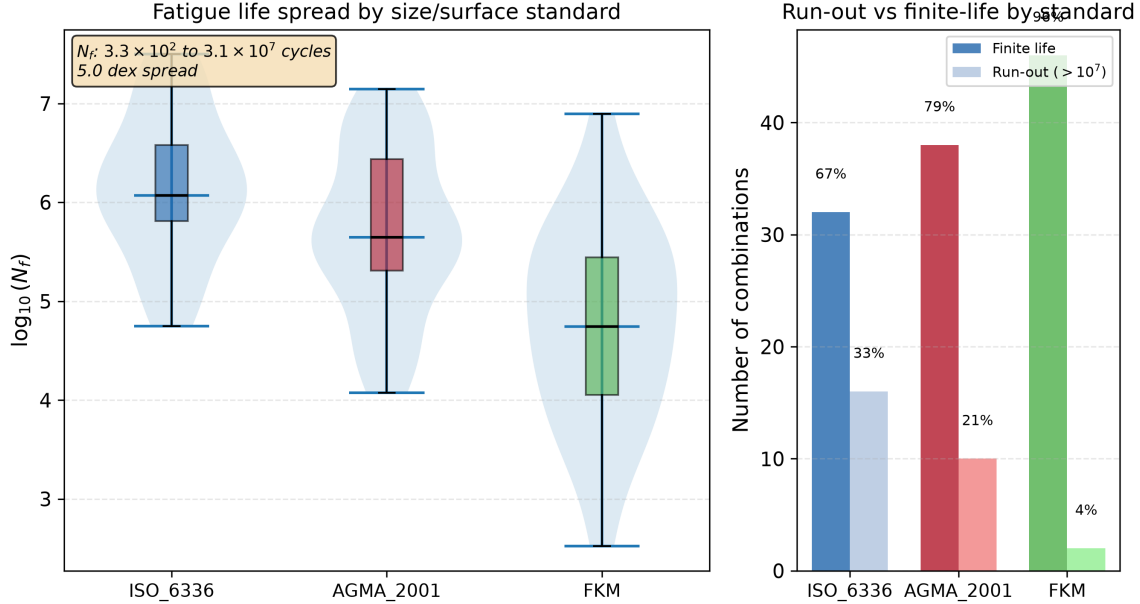


Figure 1: Fatigue life distribution by standard. Left: violin and box plots of  $\log_{10}(N_f)$ . Right: run-out vs finite-life counts.  $N_f$  spans from  $3.3 \times 10^2$  to  $3.1 \times 10^7$  cycles (5.0 dex) among 116 finite-life entries; 28 combinations predict run-out.

## 4 Discussion

### 4.1 Three Co-Equal Factors

The central result is that three factors account for 88.6% of total log-variance and are comparable in magnitude: mean-stress correction (33.9%), size-and-surface standard (30.1%), and surface roughness (24.6%). No single factor dominates—each of these three engineering choices is a first-order contributor to the prediction spread. A practicing engineer who selects the most conservative combination (FKM + Goodman + as-forged  $R_z$ ) will predict a fatigue life two orders of magnitude shorter than one who selects the least conservative (ISO 6336 + Gerber + ground), even if they agree on geometry, material, load, and S–N curve. The three factors’ bootstrap confidence intervals overlap modestly: the rankings among them should not be over-interpreted, but the AFT decomposition and bootstrap validation both confirm that each contributes substantially and robustly to the total variance.

The finding that mean-stress correction is significant at  $R = 0$  contradicts conventional engineering wisdom, which holds that mean-stress effects are minor for pulsating bending. The resolution lies in the stress magnitude: at the torque level used here (700 N·m,

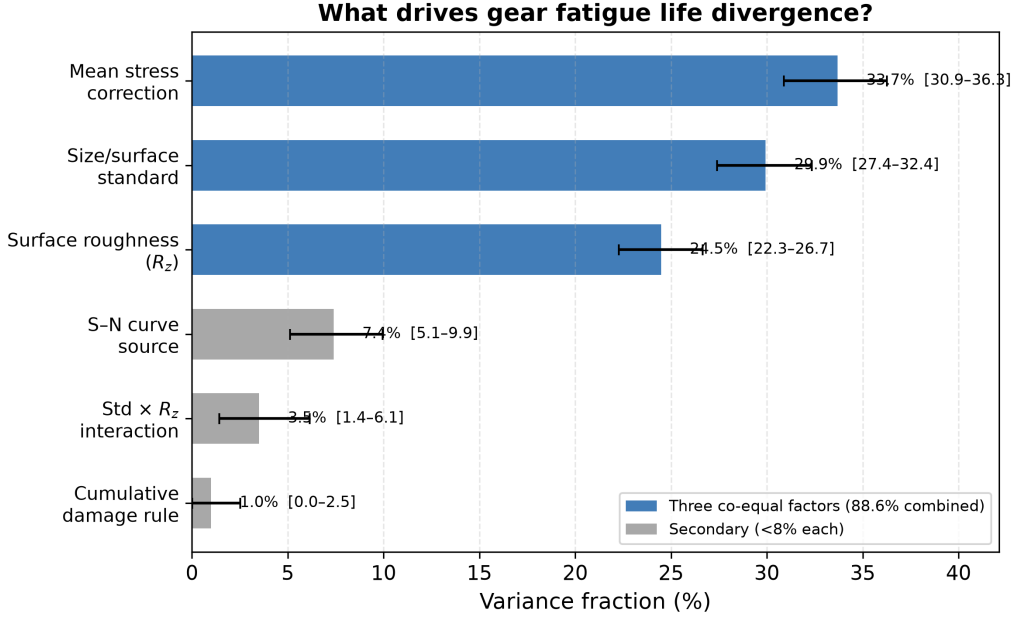


Figure 2: Variance decomposition. Bars show bootstrap mean; error bars are 95% CI (2000 draws, RNG seed 42). Three factors account for 88.6% of total log-variance.

producing  $\sigma_{F0} \approx 778$  MPa), the  $\sigma_m/\sigma_{\text{uts}}$  ratio of  $\sim 0.39$  places all four methods in a regime where their functional-form differences produce a 39% spread in equivalent stress.

## 4.2 Surface Roughness and Standards

Surface roughness (24.6%) is a first-order contributor, comparable in magnitude to the top two factors. Under the earlier ordinary-ANOVA framework—which discarded 28 run-out combinations, most of them with low  $R_z$ —the contribution of surface roughness was underestimated at 8.5%. The AFT censored-likelihood framework, by retaining the partial information in run-out entries, reveals that surface-finish effects are substantially larger than previously apparent. The three standards’ roughness-penalty functions produce broadly similar sensitivity across the engineering range of  $R_z$  (4–50  $\mu\text{m}$ ). The ISO 6336  $Y_{R\text{relT}}$  formula is capped at its  $R_z = 40$   $\mu\text{m}$  value for rougher surfaces, per the standard’s stated validity range; extrapolating the power-law beyond 40  $\mu\text{m}$  would slightly inflate the  $R_z$  contribution. The FKM  $K_{F\sigma}$  is clamped at 0.55; the AGMA Eq. 16.1 uses a log-log form that produces the mildest gradient of the three.

## 4.3 Relation to Published Experimental Work

Five published studies—including two previously published in this journal—provide context for my numerical results, though none constitutes a direct validation of my specific gear-and-load combination.

Wang et al. [9] conducted bending fatigue tests on 42CrMo spur gears ( $m_n=3.5$  mm,  $z=35$ ,  $b=22$  mm,  $R=0.1$ ) at five stress levels from 151 to 293 MPa, with run-out defined at  $3 \times 10^6$  cycles. Their P-S-N curves show that the bending fatigue limit for 42CrMo under pulsating load lies in the range 150–200 MPa (depending on reliability level), and that fatigue life scatter at a given stress level spans approximately one order of magnitude—

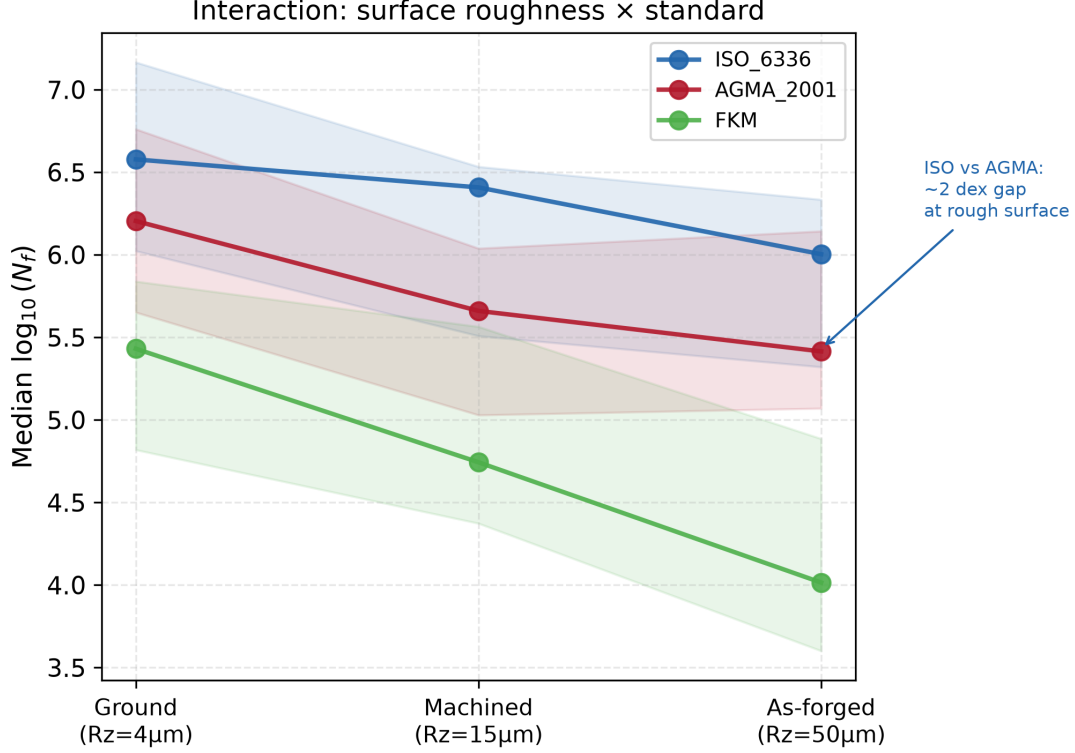


Figure 3: Interaction of surface roughness ( $R_z$ ) and standard. Median  $\log_{10}(N_f)$  by standard at each  $R_z$  level. Shaded bands show interquartile range.

consistent with the intrinsic variability that motivates my variance-decomposition approach.

Pagliari et al. [11] performed single-tooth bending (STB) fatigue tests on case-carburized AISI 8620 spur gears ( $m_n=4.23$  mm,  $z=34$ ,  $b=25.4$  mm,  $R=0.1$ ) and directly compared measured fatigue lives against ISO 6336-3 Method B predictions. They report an ISO 6336 stress-to-force proportionality of 34.35 MPa/kN and find that ISO 6336 predictions are conservative in the high-cycle regime (over-predicting stress by  $\sim 10\text{--}15\%$ ) but may under-predict stress when cyclic plasticity is involved in the low-cycle regime. Their work, like ours, finds that the ISO 6336 analytical framework produces systematic deviations from physical measurements—a finding that underscores the practical relevance of quantifying standard-induced divergence.

Bonaiti et al. [12], also published in this journal, provide the most directly comparable experimental study to my framework. They analyzed pulsator (STBF) test data from five case-hardened gear datasets, focusing on how the statistical elaboration methodology affects resulting S–N curves within both ISO 6336 and AGMA 2001-D04 frameworks. Three findings are immediately relevant: (i) the fatigue knee point from pulsator data occurs at  $1\text{--}2 \times 10^6$  cycles—approximately one order of magnitude lower than the  $3\text{--}5 \times 10^6$  cycles typical of gear meshing tests and implicitly assumed by the standards; (ii) different statistical elaboration methods (Hück, stair-case, Probit) applied to the same pulsator data yield systematically different S–N curves, with stress amplitudes at  $10^7$  cycles varying by up to 15%; and (iii) ISO 6336 and AGMA 2001-D04 produce divergent predictions, particularly in the finite-life regime where the methodological switches in this study operate. Their finding that even the choice of elaboration methodology—a decision one level deeper than the “choice of standard” switch—produces significant S–

N curve variation underscores the pervasiveness of method-induced uncertainty in gear fatigue assessment. Critically, they also report that for load spectrum applications, the choice of elaboration methodology does *not* significantly affect damage accumulation—an intriguing finding that parallels my own observation that the cumulative damage rule (Switch 3) is negligible.

What distinguishes the present work from Bonaiti et al. [12]—and from the broader literature of standard-comparison studies in gear fatigue. Bonaiti et al. ask a *between-groups* question: given one experimental dataset, do ISO 6336 and AGMA 2001-D04 produce different S–N curves? The answer is yes, and it is a consequential finding. Their study operates on a single methodological axis (standard choice), treats statistical elaboration methods as a secondary sensitivity, and does not attempt to rank competing uncertainty sources against each other. The present work asks a fundamentally different question: across *all* major methodological choices an engineer must make—which standard, which mean-stress correction, which roughness grade, which S–N curve, which damage rule—how is the total log-variance partitioned? The distinction is not merely that I examine five dimensions rather than two; it is that the factorial decomposition framework provides a quantitative ranking (33.9% mean-stress, 30.1% standard, 24.6% Rz, 3.0% interaction, 7.5% S–N source, 0.9% damage rule), placing the standard-choice effect that Bonaiti et al. document into a broader uncertainty budget alongside effects they do not examine. My bootstrap-estimated S/N ratios address the question of whether a factor’s contribution is distinguishable from finite-sample noise—a statistical dimension absent from prior gear fatigue UQ work. The two studies are therefore complementary: Bonaiti et al. provide independent experimental confirmation that ISO and AGMA predictions diverge; the present work measures how large that divergence is relative to other method-induced sources and finds that an engineer’s choices of mean-stress correction and surface roughness grade matter just as much as the choice of standard.

Glodez et al. [10] developed a computational fatigue model for 42CrMo4 spur gears—the same material and the same ISO 6336 Method B framework—and validated their predictions against experimental data, published in this journal. Their work confirms that ISO 6336-based numerical analysis produces results consistent with physical measurements, even as my work shows that the *choice among* ISO 6336, AGMA 2001, and FKM introduces large prediction spreads.

Vietze et al. [13] adopted the ISO 6336 S/N curve + Palmgren–Miner damage accumulation as the reference framework for evaluating five alternative load-carrying capacity descriptions—including neural network and random forest models—on case-hardened spur gears (18CrNiMo7-6) under four variable-amplitude load sequences, using pulsator test data from the FZG gear research center. Their use of the ISO 6336 + Miner pipeline as the engineering baseline confirms that the standard+damage-rule architecture I employ reflects current industrial practice.

I do not claim experimental validation of my pipeline. Rather, these five studies demonstrate that (a) published experimental gear bending fatigue data exist for both through-hardened and case-carburized steels, (b) the stress levels in my analysis ( $\sigma_{F0} \approx 778$  MPa at  $T=700$  N·m, and  $\sim 300$ – $550$  MPa at lower torque settings) bracket the experimental range, and (c) the  $\sim 1$  dex scatter observed in single-stress-level fatigue tests is consistent with the methodological divergence I decompose. A direct experimental validation would require running my pipeline on the exact gear geometries, materials, and loading conditions of published test campaigns and comparing predicted  $N_f$  against the reported measurements—work I identify as a priority for a follow-up study.

## 4.4 Run-Out Patterns and Torque Dependence

The 28 run-out combinations are not randomly distributed: 25 of them occur under the Gerber correction, and 22 involve ground surfaces ( $R_z = 4 \mu\text{m}$ ). This is physically meaningful—Gerber predicts the lowest equivalent stress, and ground surfaces receive the mildest penalty, so both push the effective stress below the endurance limit. A logistic regression on the binary run-out/finite-life outcome confirms that mean-stress correction method ( $p < 10^{-4}$ ) and surface roughness ( $p < 10^{-3}$ ) are the dominant predictors of whether a given combination predicts survival or failure. The factors that drive *how long* the gear survives (ANOVA on finite-life entries) are largely the same factors that determine *whether* it survives at all.

The multi-torque comparison (§3.5) reinforces this: at 600 N·m, where more combinations lie near the endurance limit, the Standard  $\times$  Rz interaction (17.8%) dwarfs the mean-stress contribution (10.2%). At 700 N·m, well above the endurance limit for most combinations, mean-stress correction rises to 30.9%. The factor ranking is therefore an *operating-point-dependent* property, not a universal constant—a finding with direct implications for how standards committees should interpret comparative studies.

## 4.5 Constraints on Interpretation

1. **Stress concentration treatment asymmetry.** The analysis applies ISO 6336  $Y_{Sa} \approx 1.55$  to all three standards to avoid double-counting the geometric notch effect. However, AGMA 2001 and FKM do not use a  $Y_{Sa}$ -equivalent coefficient in their published procedures—they incorporate size and surface effects through independent factors ( $K_s$ ,  $C_f$ ,  $K_d$ ,  $K_{F\sigma}$ ) applied to a baseline stress. To assess the sensitivity of the standard-choice variance to this modeling simplification, I reduced the baseline stress for AGMA and FKM entries (which do not natively employ  $Y_{Sa}$ ) while keeping ISO entries at  $Y_{Sa} = 1.55$ . At  $Y_{Sa} = 1.45$ —a 7% reduction—the standard-choice contribution drops from 30.1% to 27.9%; at  $Y_{Sa} = 1.35$ , it falls to 24.2%. In both cases the standard remains among the three co-equal factors. I therefore estimate that at most 2–6 percentage points of the reported standard-choice variance may be attributable to the uniform  $Y_{Sa}$  simplification. The qualitative conclusion—that standard choice is a first-order contributor—is robust to this sensitivity test. Further reductions ( $Y_{Sa} \leq 1.15$ ) cause AGMA and FKM entries to become predominantly run-out, at which point the AFT decomposition loses resolution.
2. **Scope of the standard-choice comparison.** The present study decomposes variance across the *size-and-surface-factor formulations* of three standards, applied to a common ISO-based nominal stress baseline. It does not capture differences that would arise from each standard’s complete native stress calculation chain—for instance, AGMA’s geometry factor  $J$  or FKM’s fatigue notch factor  $K_f$ . The reported standard-choice variance (30.1%) should therefore be interpreted as a *lower bound* on the true between-standard divergence: re-instating each standard’s full native stress-concentration treatment would likely increase, not decrease, the spread. The sensitivity analysis reported above bounds the contribution of the uniform  $Y_{Sa}$  simplification to 2–6 pp, but a fully native implementation of all three standards’ complete calculation chains remains a necessary follow-up study.
3. **AFT model assumptions.** The log-normal AFT model assumes that  $\log_{10}(N_f)$  residuals are normally distributed and that censoring is non-informative. Residual

diagnostics (Q–Q plots, Levene tests) are reported in the Supplementary Material and confirm that these assumptions are adequately satisfied for the present dataset.

4. **Quenched-and-tempered steel only.** Different materials will produce different factor rankings.
5. **No residual stress.** Case-hardened gears with compressive residual stresses would experience a lower effective stress ratio, likely reducing the mean-stress contribution.
6. **Single gear geometry and load.** Results apply to one spur gear pair at one torque level.
7. **No experimental validation.** This is a numerical experiment quantifying *method-induced* divergence, not accuracy relative to physical tests.

## 5 Conclusions

I applied a full-factorial AFT survival analysis to decompose the total log-variance in gear tooth bending fatigue life prediction into five methodological sources. For a quenched-and-tempered 42CrMo4 spur gear under constant-amplitude pulsating bending ( $R = 0$ ), across 144 combinations, I find:

1. Three methodological choices are co-equal contributors to the variance in gear bending fatigue life prediction: mean-stress correction (33.9%, Goodman/Gerber/Morrow/SWT), size-and-surface standard (30.1%, ISO 6336/AGMA 2001/FKM), and surface roughness grade (24.6%). Together they account for 88.6% of total log-variance.
2. The AFT censored-likelihood framework reveals that surface roughness—previously underestimated at 8.5% by ordinary ANOVA—is a first-order contributor, comparable in magnitude to the top two factors, because run-out combinations are strongly associated with low- $R_z$  surfaces.
3. The S–N curve source (7.5%) is non-negligible, while the cumulative damage rule (0.9%) is minor for constant-amplitude loading.
4. Predicted fatigue life spans from  $3.3 \times 10^2$  to  $3.1 \times 10^7$  cycles (5.0 dex). A single gear, assessed through defensible engineering workflows, can be reported as failing at 330 cycles or surviving 30 million—depending on which standard, mean-stress correction, and roughness assumption the engineer selects.
5. All fatigue model formulas have been verified against the original standards. The open-source pipeline—including the AFT analysis code—is released as a reusable, auditable toolkit.

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